Employee Attrition Prediction System using Machine Learning

ABSTRACT

This project titled **"Employee Attrition Prediction System using Machine Learning"** focuses on predicting whether an employee is likely to leave an organization based on various factors such as age, monthly income, job satisfaction, years at the company, work-life balance, and overtime.

The system is implemented using **Python, Scikit-learn, and Streamlit**, providing an interactive web-based interface where users can upload a dataset, train a machine learning model, and analyze employee attrition patterns.

The **Random Forest Classification algorithm** is used to predict employee attrition, while **K-Means Clustering and DBSCAN (Density-Based Clustering)** are used to analyze employee behavior patterns. Performance metrics such as **accuracy, precision, recall, ROC curve, and AUC** are used to evaluate the model.

The system allows users to **upload employee datasets, train models, view performance metrics, visualize clustering results, and predict employee attrition**, making it useful for HR analytics and decision-making.

INTRODUCTION

Employee attrition is a major challenge for organizations as it leads to increased recruitment costs, training expenses, and productivity loss.

Machine Learning techniques can analyze employee data and identify patterns that indicate whether an employee may leave the organization.

This project develops a **machine learning-based system that predicts employee attrition and provides insights into employee behavior using classification and clustering techniques**.

The system allows HR professionals to upload employee datasets and train models that help predict employee turnover risk. The application is built using **Streamlit**, providing an easy-to-use web interface.

OBJECTIVES

The main objectives of this project are:

- To build a machine learning model to predict employee attrition
- To allow users to upload employee datasets in CSV format
- To train the model using the **Random Forest classification algorithm**
- To evaluate model performance using **accuracy, precision, recall, ROC, and AUC**
- To analyze employee groups using **K-Means clustering**
- To identify density-based clusters using **DBSCAN clustering**
- To create an interactive web interface using **Streamlit**

SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or Laptop
- Minimum **4 GB RAM**
- Processor **Intel i3 or above**

Software Requirements

- Operating System: **Windows / Linux**
- Python **3.8 or above**
- Streamlit
- Scikit-learn
- Pandas
- NumPy
- Matplotlib

DATASET DESCRIPTION

The dataset used in this project contains information about employees and their working conditions.

The dataset is uploaded by the user in **CSV format** through the Streamlit interface.

Dataset Attributes

Attribute	Description
Age	Age of the employee
MonthlyIncome	Monthly salary of the employee
JobSatisfaction	Job satisfaction level (1–4)
YearsAtCompany	Number of years worked in the company
WorkLifeBalance	Work-life balance rating
OverTime	Whether employee works overtime (0 or 1)
Attrition	Target variable indicating if employee leaves (1) or stays (0)

TECHNOLOGIES USED

Python

Python is used as the main programming language because of its simplicity and powerful libraries for data analysis and machine learning.

Scikit-learn

Scikit-learn is used for implementing machine learning algorithms such as:

- Random Forest Classification
- K-Means Clustering
- DBSCAN Clustering
- Model evaluation metrics

Pandas

Pandas is used for:

- Reading CSV datasets
- Data preprocessing
- Data manipulation

Streamlit

Streamlit is used to create a **web-based user interface** that allows users to:

- Upload datasets
- Train models
- View performance metrics
- Predict employee attrition

MACHINE LEARNING TECHNIQUES USED

Random Forest Classification

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy.

It is used in this project to **predict employee attrition**.

Advantages:

- High accuracy
- Handles large datasets
- Reduces overfitting

K-Means Clustering

K-Means is an unsupervised learning algorithm used to group similar employees into clusters.

It helps identify **employee groups based on characteristics such as salary and experience**.

DBSCAN (Density-Based Spatial Clustering)

DBSCAN is a density-based clustering algorithm that groups points that are closely packed together.

It is useful for:

• Detecting clusters of employees
• Identifying outliers in the dataset

MODULE DESCRIPTION

Data Upload Module

• Allows users to upload employee datasets
• Reads CSV files using Pandas
• Displays dataset preview

Model Training Module

• Splits data into training and testing sets
• Trains the Random Forest model
• Generates predictions

Model Evaluation Module

Evaluates model performance using:

• Accuracy
• Precision
• Recall
• ROC Curve
• AUC Score

Clustering Module

Performs clustering using:

• K-Means clustering
• DBSCAN clustering

These algorithms help analyze employee behavior patterns.

Prediction Module

Allows users to enter employee details and predict whether the employee is likely to leave the company.

WORKING PRINCIPLE

1. The user uploads the employee dataset.

2. The dataset is read and processed using Pandas.
3. The dataset is split into training and testing data.
4. The Random Forest model is trained on the training data.
5. The model predicts employee attrition on test data.
6. Performance metrics are calculated and displayed.
7. K-Means and DBSCAN clustering are applied to analyze employee groups.
8. The user can enter employee details to predict attrition.

ALGORITHM

1. Start the application
2. Upload employee dataset
3. Preprocess dataset
4. Split dataset into training and testing data
5. Train Random Forest model
6. Predict employee attrition
7. Evaluate performance using metrics
8. Apply clustering algorithms
9. Display results and predictions

PERFORMANCE METRICS

The model is evaluated using the following metrics:

Accuracy

Measures the percentage of correct predictions.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision

Measures how many predicted positive cases are actually positive.

Precision = TP / (TP + FP)

Recall (Sensitivity)

Measures how many actual positive cases are correctly identified.

Recall = TP / (TP + FN)

ROC Curve

The ROC curve shows the trade-off between:

• True Positive Rate
• False Positive Rate

AUC (Area Under Curve)

AUC measures the overall performance of the model.

Higher AUC indicates better classification performance.

ADVANTAGES

• Helps HR departments predict employee turnover
• Reduces employee replacement cost
• Provides data-driven decision-making
• Interactive web interface
• Supports clustering analysis

APPLICATIONS

• Human Resource Analytics
• Employee retention analysis
• Workforce planning
• Organizational behavior analysis
• Talent management

LIMITATIONS

• Requires good quality dataset
• Performance depends on dataset size
• Predictions may vary with different datasets

FUTURE ENHANCEMENTS

• Integration with real HR databases
• Deep learning-based prediction models
• Cloud-based deployment
• Advanced visualization dashboards
• Automated feature selection

CONCLUSION

The **Employee Attrition Prediction System using Machine Learning** successfully demonstrates how machine learning techniques can be used to analyze employee data and predict attrition.

By combining **Random Forest classification with clustering algorithms**, the system provides both predictive and analytical insights. The use of **Streamlit** makes the application interactive and easy to use.

This project can assist HR departments in making informed decisions to reduce employee turnover and improve organizational efficiency.